① $a_n = \frac{1}{n} + \frac{1}{2n} + \frac{1}{3n} + \cdots + \frac{1}{n^2}$ —①

Let, $b_r = \frac{n}{rn}$

$b_1 + b_2 + b_3 + \cdots + b_n = \frac{n}{1\cdot n} + \frac{n}{2\cdot n} + \frac{n}{3\cdot n} + \cdots + \frac{n}{n\cdot n}$ —②

$\Rightarrow \frac{b_1 + b_2 + b_3 \cdots + b_n}{n} = \frac{1}{n} + \frac{1}{2n} + \frac{1}{3n} + \cdots + \frac{1}{n^2}$ —③

observe that, RHS of ② = RHS of ①

Now, by Cauchy's first theorem on limits we know, if $\lim_{n\to\infty} b_n = l$, then

$$\lim_{n\to\infty}\left[\frac{b_1 + b_2 + \cdots + b_n}{n}\right] = l$$

here, $b_n = \frac{n}{n^2}$ (from ②)

$\Rightarrow \lim_{n\to\infty} b_n = \lim_{n\to\infty}\left(\frac{1}{n}\right) = 0$

$\therefore \lim_{n\to\infty}\left(\frac{b_1 + b_2 + \cdots + b_n}{n}\right) = 0$

i.e. $\lim_{n\to\infty}\left(\frac{1}{n} + \frac{1}{2n} + \frac{1}{3n} + \cdots + \frac{1}{n^2}\right) = 0$ (using ③)

$\Rightarrow \boxed{\lim_{n\to\infty}(a_n) = 0}$

Alternate Solution,

Given, $a_n = \frac{1}{n} + \frac{1}{2n} + \frac{1}{3n} + \cdots + \frac{1}{n^2}$

$\Rightarrow a_n = \frac{1}{n}\left[1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}\right]$

Let, $b_n = \frac{1}{n} \Rightarrow \lim_{n\to\infty} b_n = \lim_{n\to\infty}\left(\frac{1}{n}\right) = 0$

& by Cauchy's first mean value theorem we have,

$\lim_{n\to\infty} b_n = 0$

$\Rightarrow \lim_{n\to\infty} \frac{1}{n}(b_1 + b_2 + b_3 + \cdots b_n) = 0$

$\Rightarrow \lim_{n\to\infty} \frac{1}{n}\left[1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}\right] = 0$

$\Rightarrow \lim_{n\to\infty} a_n = 0$

Q2 → Let $a_n = \left[\left(\frac{2}{1}\right)^1 \cdot \left(\frac{3}{2}\right)^2 \cdot \left(\frac{4}{3}\right)^3 \cdots \left(\frac{n+1}{n}\right)^n\right]$

$\Rightarrow a_{n+1} = \left[\left(\frac{2}{1}\right)^1\left(\frac{3}{2}\right)^2\left(\frac{4}{3}\right)^3 \cdots\cdots \left(\frac{n+1}{n}\right)^n\left(\frac{n+2}{n+1}\right)^{n+1}\right]$

$\therefore \lim_{n\to\infty}\left(\frac{a_{n+1}}{a_n}\right) = \lim_{n\to\infty}\left(\frac{n+2}{n+1}\right)^{n+1}$

$= \lim_{n\to\infty}\left[\frac{n+1+1}{n+1}\right]^{n+1}$

$= \lim_{n\to\infty}\left[1 + \frac{1}{n+1}\right]^{n+1}$

$= e$

also, $\because a_n > 0 \ \forall\ n \in \mathbb{N}$,

$\therefore$ by Cauchy's second theorem on limits, we have,

$$\lim_{n\to\infty}(a_n)^{1/n} = \lim_{n\to\infty}\left(\frac{a_{n+1}}{a_n}\right) = e$$

Q3 → Given, $a_n = \left(1 + \frac{4}{n}\right)^n$

Use binomial expansion on RHS, we get,

$$\left(1 + \frac{4}{n}\right)^n = 1 + \frac{n}{1!}\left(\frac{4}{n}\right) + \frac{n(n-1)}{2!}\left(\frac{4}{n}\right)^2 + \frac{n(n-1)(n-2)}{3!}\left(\frac{4}{n}\right)^3 + \cdots$$

$$= 1 + \frac{4}{1!} + (1 - 1/n)\cdot\frac{4^2}{2!} + \frac{(1-1/n)(1-2/n)}{3!}4^3 + \cdots$$

$$\Rightarrow \lim_{n\to\infty}\left(1 + \frac{4}{n}\right)^n = \left[1 + \frac{4}{1!} + \frac{4^2}{2!} + \frac{4^3}{3!} + \cdots\right]$$

$$= e^4$$

$$\therefore \boxed{\lim_{n\to\infty} a_n = e^4}$$